

1. Overview

ComPDF Conversion SDK is a high-performance library designed for extracting and transforming the data within your PDF files, such as text, images, tables, links, and annotations, into various file formats. The Conversion SDK retains the original document layout and the properties of the file data, helping you build a reliable document conversion workflow in C applications.

Effortlessly integrate the ComPDF Conversion SDK into your C projects in just a few steps, and enable the following file format conversions:

- Convert PDF to Word (.docx)
- Convert PDF to Excel (.xlsx)
- Convert PDF to PowerPoint (.pptx)
- Convert PDF to HTML (.html)
- Convert PDF to CSV (.csv)
- Convert PDF to Image (.png, .jpg, .jpeg, .jpeg2000, .bmp, .tiff, .tga, .gif, .webp)
- Convert PDF to Plain Text (.txt)
- Convert PDF to Rich Text Format (.rtf)
- Convert PDF to Searchable PDF (.pdf)
- Convert PDF to OFD (.ofd)
- Convert PDF to Structured Data (.json)
- Convert PDF to Markdown (.md)

To enhance format conversion results, ComPDF also provides AI-powered document tools with the following capabilities:

- Optical Character Recognition (OCR)
- Layout Analysis
- Table Recognition

1.1 Why ComPDF Conversion SDK

- Mature Technology

With years of technology accumulation, ComPDF has established a complete mechanism of product iteration to offer a continuous guarantee for product competitiveness.

- Complete PDF and Format Conversion Functionalities

The comprehensive feature set can meet diverse conversion needs and is easy for customers to use without training costs.

- High-quality Service

Professional service and technical support can quickly respond to users' feedback through onsite service or remote support such as telephone and email.

- Independent Intellectual Property Rights

The technology is independent and compliant with ISO, helping enterprises conduct international

business without copyright risks.

1.2 ComPDF Conversion SDK for C

The ComPDF Conversion SDK is designed to convert PDF files into many other formats while preserving the original layout and formatting of the documents. In this guide, we will demonstrate how to use the C API exported by the SDK in Windows, Linux, and macOS projects.

1.3 License & Trial

The ComPDF Conversion SDK is a commercial SDK that requires a license to grant developers the right to develop and distribute their applications. In development mode, each license is only valid for one device ID. ComPDF provides flexible licensing models. Please contact [our marketing team](#) for more information. Even if you have a license, it is prohibited to distribute any documents, sample code, or source code of the ComPDF Conversion SDK to any third parties.

If you do not have a license, please contact the [ComPDF Team](#) to obtain a trial license for ComPDF Conversion SDK.

2. Get Started

2.1 Requirements

Before starting, please make sure that you have met the following prerequisites.

2.1.1 Get ComPDF License Key

ComPDF provides two types of license key: 30-day free trial license and commercial license.

How to Get Free Trial License

[Contact our sales team](#) and we will send you a 30-day free trial license for ComPDF Conversion SDK.

How to Get Commercial License

ComPDF Conversion SDK is a commercial SDK that requires a license for application release. Any documents, sample code, or source code distribution from the released package of ComPDF to any third party is prohibited.

To get a commercial license for ComPDF Conversion SDK, feel free to [contact our sales team](#).

For the C Conversion SDK, the commercial license must be bound to your developer device ID ([How to find the developer device ID](#)), and each license is only valid for one device ID in development mode.

2.1.2 Download Conversion SDK

[Contact us](#) to obtain the ComPDF C Conversion SDK.

2.1.3 System Requirements

Development Platform	System Requirements	Development Environment	Notice
Windows	- Windows 7, 8, 10, and 11 (32-bit, 64-bit). - A C compiler compatible with the delivered library architecture.	Visual Studio 2022 or higher.	Samples have been tested on Windows 10 and 11.
Linux	- Linux x64.	GCC or Clang toolchain.	Samples have been tested on Ubuntu 20.04.
Mac	- macOS 10.14 or higher (Intel, Apple Silicon).	Xcode 13.0 or higher.	Samples have been tested on Mac Intel and Mac Apple Silicon.

2.2 SDK Package Structure

You can [contact us](#) to get the PDF format conversion SDK package. The ComPDF Conversion C SDK contains the following files:

- **"doc"** - API reference and developer guide.
- **"include"** - Header files for ComPDF Conversion SDK C API.
- **"lib"** - ComPDF dynamic libraries for supported architectures.
- **"resource"** - DocumentAI model resources.
- **"samples"** - Sample projects and scripts.
- **"legal.txt"** - Legal and copyright information.
- **"release_notes.txt"** - Release information.



2.3 Apply the License Key

If you do not have a license key, please check [how to obtain a license key](#).

ComPDF Conversion SDK currently supports offline authentication to verify license keys.

Learn about:

[What is the authentication mechanism of ComPDF's license?](#)

2.3.1 Copy the License Key

Accurately obtaining the license key is crucial for applying the license.

1. In the email you received, locate the XML file containing the license key.
2. Open the XML file and determine the license type based on the `<type>` field. If `<type>online</type>` is present, it indicates an online license. If `<type>offline</type>` is present or if the field is absent, it indicates an offline license.

Online License:

```
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<license version="1">
  <platform>windows</platform>
  <starttime>xxxxxxx</starttime>
  <endtime>xxxxxxx</endtime>
  <type>online</type>
  <key>LICENSE_KEY</key>
</license>
```

Offline License:

```
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<license version="1">
  <platform>windows</platform>
  <starttime>xxxxxxx</starttime>
  <endtime>xxxxxxx</endtime>
  <key>LICENSE_KEY</key>
</license>
```

3. Copy the value located at the `LICENSE_KEY` position within the `<key>LICENSE_KEY</key>` field. This is your license key.

2.3.2 Apply the License Key

You can perform offline authentication using the following method:

```
#include "compdf_common_c.h"

const char* license = "LICENSE_KEY";
CSDKErrorCode code = CPDF_LicenseVerify(license, "device_id", "app_id");
if (code != e_CErrSuccess) {
    return code;
}
```

Before calling any conversion API, initialize the SDK resource directory:

```
#include "compdf_common_c.h"

#if defined(_WIN32)
#define CPDF_TEXT(text) L##text
#else
#define CPDF_TEXT(text) text
#endif

CPDF_Initialize(CPDF_TEXT("ComPDF_Conversion_SDK/resource"));
```

2.4 How to Run a Demo

2.4.1 Windows

ComPDF Conversion SDK provides a demo in the **"samples"** folder. You can follow these steps to run the demo:

1. Load the Visual Studio solution file **"demo_vs2022.sln"** in the **"samples"** folder.
2. Build the demo by clicking **Build > Build Solution**. After the build is completed, the executable file **".exe"** will be generated in the **"samples/bin"** folder. The executable file name depends on the build configuration.
3. Double-click the executable file to run the demo.

Output files such as Word, Excel, and PowerPoint files will be generated in the **"samples/output_files"** folder.

2.4.2 Linux and Mac

ComPDF Conversion SDK provides demos in the **"samples"** folder. Before running the demo, make sure you have configured your environment correctly and installed CMake 3.0 or higher. To run the demo, follow these steps:

1. Open a terminal window and navigate to the **"samples"** folder of the ComPDF Conversion SDK package.
2. Enter the following command to run the demo.

```
./RunDemo.sh
```

Output files such as Word, Excel, and PowerPoint files will be generated in the **"samples/output_files"** folder.

3. C API Basics

3.1 Header Files

The C SDK exposes the following main headers:

Header	Description
<code>compdf_common_c.h</code>	License, initialization, resource release, page count, version, and DocumentAI model configuration.

Header	Description
<code>compdf_conversion.c.h</code>	PDF conversion APIs.
<code>compdf_basictypes.c.h</code>	Error codes, conversion options, image types, OCR languages, and other public C types.

Most conversion code needs at least:

```
#include <stdbool.h>
#include <string.h>

#include "compdf_common.c.h"
#include "compdf_conversion.c.h"
```

3.2 Platform String Type

Most file path parameters use `COMPDFKIT_STRING`:

```
#if defined(_WIN32)
#define COMPDFKIT_STRING const wchar_t*
#else
#define COMPDFKIT_STRING const char*
#endif
```

On Windows, pass wide-character strings such as `L"input.pdf"`. On Linux and macOS, pass UTF-8 `const char*` strings. For portable string literals in sample code, this guide uses the following helper macro:

```
#if defined(_WIN32)
#define CPDF_TEXT(text) L##text
#else
#define CPDF_TEXT(text) text
#endif
```

`CPDF_LicenseVerify` is an exception: its `license`, `device_id`, and `app_id` parameters are always `const char*`.

3.3 Conversion Option Initialization

`CConvertOption` is a plain C structure. It does not have a constructor, so initialize it before use. The following helper mirrors the current default values used by the C++ SDK:

```
#include <string.h>

static CConvertOption CPDF_DefaultConvertOption(void)
{
    CConvertOption option;
    memset(&option, 0, sizeof(option));

    option.enable_ai_layout = true;
```

```

option.enable_ai_table_recognition = true;
option.contain_image = true;
option.contain_page_background_image = true;
option.json_contain_table = true;
option.contain_annotation = true;
option.transparent_text = true;
option.txt_table_format = true;
option.formula_to_image = true;
option.auto_create_folder = true;
option.image_scaling = 4.0f;
option.page_layout_mode = e_PageLayoutModeFlow;
option.excel_worksheet_option = e_CForTable;
option.html_option = e_CSinglePage;
option.ocr_option = e_Call;
option.image_color_mode = e_CColor;
option.image_type = e_CJPG;

return option;
}

```

You can also zero-initialize the structure and explicitly set every option you rely on:

```

CConvertOption option = {0};
option.contain_image = true;
option.page_layout_mode = e_PageLayoutModeFlow;

```

For string fields such as `page_ranges` and `font_name`, use bounded copy functions in production code. The sample snippets use `strcpy` only for clarity.

3.4 Error Codes

Most C APIs return `CSDKErrorCode`. `e_CErrSuccess` indicates success. Other common error codes include:

Error Code	Meaning
<code>e_CErrFile</code>	File cannot be found or opened.
<code>e_CErrPDFPassword</code>	The PDF password is invalid.
<code>e_CErrPDFFormat</code>	The PDF format is invalid or the file is corrupted.
<code>e_CErrOutOfMemory</code>	Memory allocation failed.
<code>e_CErrLicenseInvalid</code>	The license is invalid.
<code>e_CErrLicenseExpire</code>	The license has expired.
<code>e_CErrLicenseConcurrencyExceeded</code>	Current conversion concurrency exceeds the license limit.
<code>e_CErrLicensePageLimitExceeded</code>	Current conversion pages exceed the license limit.
<code>e_CErrOCRFailure</code>	OCR recognition failed.

Error Code	Meaning
<code>e_CErrConverting</code>	A conversion task is already running.
<code>e_CErrUnknown</code>	Unknown error.

Always check the returned error code after calling a conversion API.

4. Conversion Guides

ComPDF Conversion SDK allows developers to use simple C APIs to convert PDFs to common formats such as Word, Excel, PowerPoint, HTML, CSV, PNG, JPEG, RTF, TXT, Searchable PDF, OFD, JSON, and Markdown. It also provides conversion options, such as whether to include images or annotations, whether to enable OCR, and whether to enable layout analysis.

4.1 Initialize Library Resources

Overview

Initialize the necessary file and memory resources required by the ComPDF Conversion SDK.

Notes

- You must initialize SDK resources before calling any conversion interface.
- When using OCR, Layout Analysis, Table Recognition, PDF to Searchable PDF, or PDF to OFD, make sure the DocumentAI model resources in the `resource` directory are available.

Example

```
CPDF_Initialize(CPDF_TEXT("ComPDF_Conversion_SDK/resource"));
```

4.2 Set DocumentAI Model

Overview

Before using OCR, Layout Analysis, Table Recognition, PDF to Searchable PDF, or PDF to OFD, set the DocumentAI model path first.

`CPDF_SetDocumentAIModel` supports the `gpu_id` parameter to specify the GPU device index for the AI model. When `gpu_id` is `-1`, GPU acceleration is disabled.

Example

```
CSDKErrorCode code = CPDF_SetDocumentAIModel(CPDF_TEXT("path/documentai.model"), -1);
if (code != e_CErrSuccess) {
    return code;
}
```


Set AI Model Instance Count

If you need to control the number of Layout Analysis and Table Recognition model instances, call the following interface:

```
CPDF_SetDocumentAIModelCount(1, 1);
```

The first parameter indicates the number of Layout Analysis model instances, and the second parameter indicates the number of Table Recognition model instances.

Use Your Own AI Engine (SDK v4.1.0+)

This option is available only in SDK v4.1.0 or later. If you prefer to run OCR, Layout Analysis, or Table Recognition with your own model or a third-party service instead of the bundled DocumentAI model, the SDK exposes callback hooks on `CConvertCallback` that let you supply the results as JSON. See [4.11 Use Custom AI Models via Callbacks](#) for details. When all capabilities you need are covered by your own callbacks,

`CPDF_SetDocumentAIModel` does not have to be called.

4.3 Get Conversion Progress

ComPDF Conversion SDK obtains conversion progress through the `progress` callback in `CConvertCallback`. The following example demonstrates how to get conversion progress while performing a PDF to Word task:

```
#include <stdio.h>

static void Progress(int current_page_count, int total_page_count)
{
    printf("progress: %d / %d\n", current_page_count, total_page_count);
}

CConvertOption option = CPDF_DefaultConvertOption();

CConvertCallback callback = {0};
callback.progress = Progress;

CSDKErrorCode code = CPDF_StartPDFToWord(
    CPDF_TEXT("input.pdf"),
    CPDF_TEXT("password"),
    CPDF_TEXT("path/output.docx"),
    option,
    &callback);
```

The `handle` field is maintained internally by the SDK during conversion, so it should be initialized to `NULL` when passed in.

4.4 Cancel Conversion Task

ComPDF Conversion SDK supports interrupting an ongoing conversion task through the `cancel` callback in `CConvertCallback`. When the `cancel` callback returns `true`, the current conversion task stops as soon as possible.

```
static bool Cancel(void)
{
    return false;
}

CConvertOption option = CPDF_DefaultConvertOption();

CConvertCallback callback = {0};
callback.cancel = Cancel;

CSDKErrorCode code = CPDF_StartPDFToWord(
    CPDF_TEXT("input.pdf"),
    CPDF_TEXT("password"),
    CPDF_TEXT("path/output.docx"),
    option,
    &callback);
```

If you need to cancel the conversion at a specific time, return `true` from `Cancel` based on external state.

4.5 Select Page Range for Conversion

ComPDF Conversion SDK supports converting a specified page range. When an empty string is passed, all pages will be converted. If the page range exceeds one page, you can enable `output_document_per_page` to output each PDF page as a separate file.

```
CConvertOption option = CPDF_DefaultConvertOption();
option.output_document_per_page = true;
strcpy(option.page_ranges, "1-3,5,7-9");
```

4.6 Contain Image and Annotation Options

Overview

When converting PDF documents into various formats, ComPDF Conversion SDK offers two common options: whether images are included in the generated document, and whether annotations from the PDF file are retained.

- When `contain_image` is enabled, the SDK extracts images from the PDF document and embeds them in the corresponding pages and positions in the output file. For areas with overlapping images, the SDK merges these images into one image and embeds it at the correct location.
- When `contain_annotation` is enabled, most annotations are converted into raster images and embedded at the corresponding positions. Certain types of annotations, such as highlights, underlines, strikeouts, and squiggly lines, are converted into native formatting equivalents in Word, PowerPoint, and HTML documents when possible.

These options are commonly used in the following conversions:

- PDF to Word
- PDF to Excel
- PDF to PowerPoint

- PDF to HTML
- PDF to RTF
- Extract PDF to JSON
- Extract PDF to Markdown

Sample

```
CConvertOption option = CPDF_DefaultConvertOption();
option.contain_image = true;
option.contain_annotation = true;

CSDKErrorCode code = CPDF_StartPDFToWord(
    CPDF_TEXT("input.pdf"),
    CPDF_TEXT("password"),
    CPDF_TEXT("path/output.docx"),
    option,
    NULL);
```

4.7 Page Layout Mode

In certain formats, the page layout mode plays a key role in the quality of the converted document. ComPDF Conversion SDK supports two layout modes: Flow Layout and Box Layout.

- **Flow Layout:** This layout uses paragraph indentations, columns, and tab positions to adjust content. Its main advantage is flexibility. Content can flow automatically as the document is edited and can adapt to different screen sizes.
- **Box Layout:** This layout is based on the PDF fixed-page model and accurately positions text, images, and tables on the page using coordinates. It is useful for documents that require high-precision reproduction, such as contracts, design drafts, and academic papers.

Page layout modes are commonly used in the following conversions:

- PDF to Word
- PDF to HTML

Sample

```
CConvertOption option = CPDF_DefaultConvertOption();

option.page_layout_mode = e_PageLayoutModeFlow;
CPDF_StartPDFToWord(CPDF_TEXT("input.pdf"), CPDF_TEXT("password"),
    CPDF_TEXT("path/flow.docx"), option, NULL);

option.page_layout_mode = e_PageLayoutModeBox;
CPDF_StartPDFToWord(CPDF_TEXT("input.pdf"), CPDF_TEXT("password"),
    CPDF_TEXT("path/box.docx"), option, NULL);
```

4.8 OCR

Overview

OCR (Optical Character Recognition) converts images of typed, handwritten, or printed text into machine-encoded text.

OCR is commonly used for text recognition and extraction from the following types of documents:

- Non-editable scanned PDF files.
- Photographs of documents.
- Scene photos such as advertising layouts and signboards.
- Identification cards, passports, vehicle license plates, invoices, bills, and receipts.

The following features support OCR:

- PDF to Word
- PDF to Excel
- PDF to PowerPoint
- PDF to HTML
- PDF to RTF
- PDF to TXT
- PDF to CSV
- PDF to Searchable PDF
- PDF to OFD
- Extract PDF to JSON
- Extract PDF to Markdown

OCR Language

Pass OCR languages through `CConvertOption.languages` and set `CConvertOption.language_count` for each conversion task.

```
COCRLanguage languages[] = {e_CENGLISH};

CConvertOption option = CPDF_DefaultConvertOption();
option.enable_ocr = true;
option.languages = languages;
option.language_count = 1;

CPDF_StartPDFToWord(CPDF_TEXT("word.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.docx"), option, NULL);
```

Supported C enum values include:

Enum	Description
<code>e_CCHINESE</code>	Chinese Simplified
<code>e_CCHINESE_TRA</code>	Chinese Traditional
<code>e_CENGLISH</code>	English
<code>e_CKOREAN</code>	Korean
<code>e_CJAPANESE</code>	Japanese
<code>e_CLATIN</code>	Latin script languages
<code>e_CDEVANAGARI</code>	Devanagari
<code>e_CCYRILLIC</code>	Cyrillic
<code>e_CARABIC</code>	Arabic
<code>e_CTAMIL</code>	Tamil
<code>e_CTELUGU</code>	Telugu
<code>e_CKANNADA</code>	Kannada
<code>e_CTHAI</code>	Thai
<code>e_CGREEK</code>	Greek
<code>e_CEs1av</code>	Eslav
<code>e_CAUTO</code>	Automatically select language

OCR Options

Different OCR options can be selected according to actual needs:

- `e_CInvalidcharacter`: Recognizes invalid or garbled characters in the PDF document through OCR, while normal characters are not processed by OCR.
- `e_CScanPage`: Recognizes scanned pages in the PDF document through OCR, while editable pages are not processed by OCR.
- `e_CInvalidCharacterAndScanPage`: Recognizes both invalid characters and scanned pages in the PDF document through OCR.
- `e_CAll`: Recognizes all pages and characters in the PDF document through OCR.

Preserve Page Background

When OCR is enabled, you can enable `contain_page_background_image` to preserve the original page background image of the PDF. If it is disabled, the image result detected during page layout analysis will be retained.

Notice

- The quality of the OCR result depends on the quality of the input image. A good rule of thumb is that the more pixels in the character shapes, the better. The ideal image is a grayscale image with a resolution around 300 DPI.
- When performing OCR, make sure the OCR language setting matches the language in the PDF document to achieve the best OCR conversion quality.
- OCR functionality currently does not support operating systems lower than Windows 10.

Converting Images to Other Document Formats

The OCR function also supports converting input images into Word, Excel, PowerPoint, HTML, CSV, RTF, TXT, JSON, and other formats.

```
CPDF_SetDocumentAIModel(CPDF_TEXT("path/documentai.model"), -1);

COCRLanguage languages[] = {e_CENGLISH};

CConvertOption option = CPDF_DefaultConvertOption();
option.enable_ocr = true;
option.languages = languages;
option.language_count = 1;

CPDF_StartPDFToWord(CPDF_TEXT("input.png"), CPDF_TEXT(""), CPDF_TEXT("path/output.docx"),
option, NULL);
```

4.9 Layout Analysis

Overview

Layout analysis uses AI technology to parse and understand the structure of a document layout. It extracts text, images, tables, layers, and other data from input documents.

Features that support Layout Analysis:

- PDF to Word
- PDF to Excel
- PDF to PowerPoint
- PDF to HTML
- PDF to RTF
- PDF to TXT
- PDF to CSV
- Extract PDF to JSON
- Extract PDF to Markdown

Notice

- You need to load the DocumentAI model before using layout analysis, or plug in your own AI engine via callbacks described in [4.11 Use Custom AI Models via Callbacks](#).
- When OCR is enabled, layout analysis is automatically enabled.
- AI table recognition is a separate stage controlled by `enable_ai_table_recognition`.

Sample

```
CPDF_SetDocumentAIModel(CPDF_TEXT("path/documentai.model"), -1);

CConvertOption option = CPDF_DefaultConvertOption();
option.enable_ai_layout = true;

CPDF_StartPDFToWord(CPDF_TEXT("word.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.docx"), option, NULL);
```

4.10 Table Recognition

Overview

Table Recognition reconstructs the internal structure of tables detected during layout analysis, including rows, columns, merged cells, and cell boundaries, so that the converted document preserves the original tabular semantics instead of producing a flat grid of text fragments.

It is controlled by the independent option `enable_ai_table_recognition`, which is enabled by default in the helper in this guide. The table model is invoked for table regions detected by layout analysis.

Typical scenarios that benefit from Table Recognition:

- Borderless or partially bordered tables.
- Tables with merged header cells, multi-row headers, or spanning cells.
- Scanned tables processed by OCR.

Notice

- Table Recognition runs only when layout analysis is active, either through `enable_ai_layout = true` or implicitly through `enable_ocr = true`.
- You need to load the DocumentAI model before using Table Recognition, or plug in your own table model via callbacks described in [4.11 Use Custom AI Models via Callbacks](#).
- Set `enable_ai_table_recognition = false` to disable the table model.
- The number of Table Recognition model instances can be tuned through the second parameter of `CPDF_SetDocumentAIModelCount`.

Sample

```
CPDF_SetDocumentAIModel(CPDF_TEXT("path/documentai.model"), -1);

CConvertOption option = CPDF_DefaultConvertOption();
option.enable_ai_layout = true;
option.enable_ai_table_recognition = true;

CPDF_StartPDFToWord(CPDF_TEXT("word.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.docx"), option, NULL);
```

4.11 Use Custom AI Models via Callbacks

Overview

Starting with SDK v4.1.0, ComPDF Conversion SDK exposes a callback-based extension point that lets you plug in your own AI inference engine for OCR, Layout Analysis, and Table Recognition. Instead of relying on the built-in DocumentAI model loaded by `CPDF_SetDocumentAIModel`, you can run inference with any model or service and return the result to the SDK as a JSON string.

When the relevant callback pair is registered on `cconvertcallback`, the SDK skips its built-in model invocation for that capability and consumes your JSON output instead. If a pair is left as `NULL`, the SDK falls back to the built-in DocumentAI model when available.

Callback Pairs

Each AI capability uses two callbacks: a trigger callback invoked by the SDK with the path to a page image saved as PNG in a temporary directory, and a result getter invoked by the SDK immediately afterwards to retrieve the JSON string.

Capability	Trigger callback	Result getter callback	Triggered when
OCR	<code>ocr</code>	<code>get_ocr_result</code>	<code>enable_ocr = true</code>
Layout Analysis	<code>layout</code>	<code>get_layout_result</code>	<code>enable_ai_layout = true</code> or <code>enable_ocr = true</code>
Table Recognition	<code>table</code>	<code>get_table_result</code>	<code>enable_ai_table_recognition = true</code> and a table region is detected by layout analysis

Rules:

- The trigger receives a UTF-8 path to a PNG file. Return `true` if inference succeeded, or `false` to make the SDK ignore the result for that page.
- The getter must return a UTF-8 JSON C string. The pointer must remain valid until the SDK has finished parsing it, usually until the next call to the same getter.
- Both callbacks for a capability must be set together. If only one is provided, the SDK falls back to the built-in path.

- Coordinates in your JSON must be in the pixel space of the image received by the trigger, with top-left origin, X to the right, and Y down.

Sample

```
#include <stdbool.h>
#include <stddef.h>

#include "compdf_common_c.h"
#include "compdf_conversion_c.h"

static char g_ocr_json[4096];
static char g_layout_json[4096];
static char g_table_json[4096];

static bool MyOcrTrigger(const char* image_path)
{
    return RunMyOcrModel(image_path, g_ocr_json, sizeof(g_ocr_json));
}

static const char* MyOcrGetter(void)
{
    return g_ocr_json;
}

static bool MyLayoutTrigger(const char* image_path)
{
    return RunMyLayoutModel(image_path, g_layout_json, sizeof(g_layout_json));
}

static const char* MyLayoutGetter(void)
{
    return g_layout_json;
}

static bool MyTableTrigger(const char* image_path)
{
    return RunMyTableModel(image_path, g_table_json, sizeof(g_table_json));
}

static const char* MyTableGetter(void)
{
    return g_table_json;
}

int main(void)
{
    CPDF_LicenseVerify("LICENSE_KEY", "device_id", "app_id");
    CPDF_Initialize(CPDF_TEXT("resource"));

    CConvertCallback callback = {0};
    callback.ocr = MyOcrTrigger;
```

```

callback.get_ocr_result = MyOcrGetter;
callback.layout = MyLayoutTrigger;
callback.get_layout_result = MyLayoutGetter;
callback.table = MyTableTrigger;
callback.get_table_result = MyTableGetter;

COCRLanguage languages[] = {e_CENGLISH};

CConvertOption option = CPDF_DefaultConvertOption();
option.enable_ocr = true;
option.enable_ai_layout = true;
option.languages = languages;
option.language_count = 1;

CPDF_StartPDFToWord(CPDF_TEXT("input.pdf"), CPDF_TEXT(""), CPDF_TEXT("output.docx"),
option, &callback);
CPDF_Release();
return 0;
}

```

You can register only the capabilities you want to override and leave others as `NULL` to keep the built-in behavior.

Thread Safety and Lifetime

- Callbacks are invoked from the SDK conversion thread. Implement them in a thread-safe manner if your model engine is shared across calls.
- The PNG image at `image_path` lives in the SDK temporary directory and may be deleted shortly after the trigger returns. Copy or process it before returning.
- The JSON pointer returned from a getter must remain valid until the same getter is called again or until the conversion task ends.

OCR Result JSON Schema

Returned by `get_ocr_result`. The SDK populates each `text_spans[].chars[]` either from `words[]` if provided, or by uniformly splitting the span rect.

```

{
  "text_spans": [
    {
      "text": "Hello world",
      "confidence": 0.98,
      "rotation": 0.0,
      "rect": { "left": 120, "top": 80, "right": 320, "bottom": 110 },
      "style": {
        "font_size": 18.0,
        "font_color": { "r": 0, "g": 0, "b": 0 }
      },
      "words": [
        { "text": "Hello", "rect": { "left": 120, "top": 80, "right": 200, "bottom": 110 }
      ]
    }
  ],
}

```

```

    { "text": "world", "rect": { "left": 210, "top": 80, "right": 320, "bottom": 110 } }
  ]
}
]
}

```

Field	Type	Required	Description
<code>text_spans</code>	array	Yes	Recognized text spans on the page.
<code>text</code>	string	Yes	UTF-8 text content of the span.
<code>confidence</code>	number	No	0.0 – 1.0. Spans below 0.1 are discarded.
<code>rotation</code>	number	No	Text rotation in degrees. Default 0.
<code>rect</code>	object	Yes	Bounding box in image pixels (<code>left</code> / <code>top</code> / <code>right</code> / <code>bottom</code>).
<code>style.font_size</code>	number	No	Estimated font size in pixels.
<code>style.font_color</code>	object	No	{ <code>r</code> , <code>g</code> , <code>b</code> } 0 – 255.
<code>words</code>	array	No	Per-word boxes. If omitted, the SDK splits the span rect evenly. Strongly recommended for CJK + Latin mixed lines for correct glyph spacing.

Layout Analysis Result JSON Schema

Returned by `get_layout_result`. Objects with `confidence < 0.45` are discarded.

```

{
  "objects": [
    { "type": "title", "confidence": 0.95, "rect": { "left": 60, "top": 50, "right": 540, "bottom": 90 } },
    { "type": "paragraph", "confidence": 0.97, "rect": { "left": 60, "top": 100, "right": 540, "bottom": 220 } },
    { "type": "figure", "confidence": 0.92, "rect": { "left": 80, "top": 240, "right": 520, "bottom": 460 } },
    { "type": "table", "confidence": 0.93, "rect": { "left": 60, "top": 480, "right": 540, "bottom": 700 } }
  ]
}

```

Supported `type` values:

Value	Meaning
<code>paragraph</code>	Body text paragraph
<code>title</code>	Heading

Value	Meaning
<code>figure</code>	Image or figure
<code>figure_title</code>	Figure caption header
<code>figure_caption</code>	Figure caption text
<code>table</code>	Table region. Whether the table is bordered or borderless is determined by the table recognition stage, not by the layout label.
<code>table_title</code>	Table caption header
<code>table_caption</code>	Table caption text
<code>ordered_list</code>	Ordered list
<code>unordered_list</code>	Unordered list
<code>catalogue</code>	Table of contents
<code>formula</code>	Math formula
<code>code</code>	Code block
<code>algorithm</code>	Algorithm block
<code>header</code>	Page header
<code>footer</code>	Page footer
<code>page_number</code>	Page number
<code>reference</code>	Reference or citation

Objects with a `type` value that is not listed above are ignored. Use the values in this table as the canonical layout labels in your custom output.

Table Recognition Result JSON Schema

Returned by `get_table_result` once per detected table region. Polygons use eight integers `[x0, y0, x1, y1, x2, y2, x3, y3]` in the order top-left, top-right, bottom-right, bottom-left.

```
{
  "type": "table_with_line",
  "position": [60, 480, 540, 480, 540, 700, 60, 700],
  "rows": 3,
  "cols": 2,
  "angle": 0.0,
  "height_of_rows": [40, 60, 60],
  "width_of_cols": [200, 280],
  "table_cells": [
    {
      "start_row": 0,
```

```

    "end_row": 0,
    "start_col": 0,
    "end_col": 0,
    "cell_background_color_r": 240,
    "cell_background_color_g": 240,
    "cell_background_color_b": 240,
    "position": [60, 480, 260, 480, 260, 520, 60, 520]
  }
]
}

```

Field	Type	Description
<code>type</code>	string	<code>table_with_line</code> for bordered tables; any other value is treated as a non-standard (borderless) table.
<code>position</code>	int[8]	Table polygon in image pixels.
<code>rows</code> / <code>cols</code>	int	Row / column counts.
<code>angle</code>	number	Skew angle in degrees.
<code>height_of_rows</code>	int[]	Per-row pixel heights (length = <code>rows</code>).
<code>width_of_cols</code>	int[]	Per-column pixel widths (length = <code>cols</code>).
<code>table_cells[]</code>	array	One entry per merged cell.
<code>start_row</code> / <code>end_row</code>	int	Inclusive row span of the cell.
<code>start_col</code> / <code>end_col</code>	int	Inclusive column span of the cell.
<code>cell_background_color_*</code>	int	Cell background color components (0 – 255).
<code>position</code>	int[8]	Cell polygon in image pixels.

Tip: Validate Your JSON

If you need a reference output to compare against, run a conversion once with the built-in DocumentAI model. The SDK uses the same JSON shape internally, so your custom output should follow the same structure.

4.12 Output Font Option

Overview

In some output formats, you can set the preferred font name to unify the default font style in the output document.

Supported Formats

The `font_name` option currently applies to the following formats:

- PDF to Word
- PDF to Excel
- PDF to PowerPoint
- PDF to Searchable PDF
- PDF to OFD

For Searchable PDF and OFD, `font_name` controls the font used for the invisible or visible text layer that is overlaid on the page background. For Word, Excel, and PowerPoint, it sets the preferred default font of the generated document.

Example

```
CConvertOption option = CPDF_DefaultConvertOption();
strcpy(option.font_name, "Arial");

CPDF_StartPDFToWord(CPDF_TEXT("word.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.docx"), option, NULL);
```

4.13 Convert PDF to Word

Overview

Converting PDF to Word converts a PDF file into an editable Word file. You can edit, modify, insert, or delete text and pictures, and adjust layout and properties.

Sample

```
CConvertOption option = CPDF_DefaultConvertOption();

option.page_layout_mode = e_PageLayoutModeBox;
CPDF_StartPDFToWord(CPDF_TEXT("word.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output_box.docx"), option, NULL);

option.page_layout_mode = e_PageLayoutModeFlow;
CPDF_StartPDFToWord(CPDF_TEXT("word.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output_flow.docx"), option, NULL);
```

Convert Formulas to Images

When a document contains complex formulas and you want to preserve visual consistency in the output document, enable `formula_to_image`.

```
CConvertOption option = CPDF_DefaultConvertOption();
option.formula_to_image = true;

CPDF_StartPDFToWord(CPDF_TEXT("word.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.docx"), option, NULL);
```

4.14 Convert PDF to Excel

Overview

ComPDF Conversion SDK supports converting PDF documents to Microsoft Excel format (.xlsx). By extracting, parsing, and importing data from PDF into Excel, users can further edit, analyze, or share Excel files.

Excel Options

When converting PDF files to Excel files, pay attention to the following options:

- `excel_all_content`: If enabled, the converted XLSX file contains all content in the PDF.
- `excel_worksheet_option`: Controls how worksheets are created.

Option	Description
<code>e_CForTable</code>	Create one sheet for one table.
<code>e_CForPage</code>	Create one sheet for one PDF page.
<code>e_CForDocument</code>	Create one sheet for the entire PDF document.

Sample

```
CConvertOption option = CPDF_DefaultConvertOption();
CPDF_StartPDFToExcel(CPDF_TEXT("excel.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.xlsx"), option, NULL);

option.excel_all_content = true;
option.excel_worksheet_option = e_CForDocument;
CPDF_StartPDFToExcel(CPDF_TEXT("excel.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output_all.xlsx"), option, NULL);
```

4.15 Convert PDF to PowerPoint

Overview

ComPDF Conversion SDK converts PDF files to PowerPoint files and restores the layout and format of the original document for presentation and editing in Microsoft PowerPoint.

Sample

```
CConvertOption option = CPDF_DefaultConvertOption();
CPDF_StartPDFToPpt(CPDF_TEXT("ppt.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.pptx"), option, NULL);
```

4.16 Convert PDF to HTML

Overview

ComPDF Conversion SDK converts PDF files to HTML files while maintaining the layout and format of the original document, allowing users to browse and view the document on the web.

HTML Options

Option	Description
<code>e_CSinglPage</code>	Convert the entire PDF file into a single HTML file.
<code>e_CSinglPagewithBookmark</code>	Convert the PDF file into a single HTML file with an outline for navigation at the beginning of the HTML page.
<code>e_CMulTiPage</code>	Convert the PDF file into multiple HTML files.
<code>e_CMulTiPagewithBookmark</code>	Convert the PDF file into multiple HTML files with an outline HTML file for navigation.

Sample

```
CConvertOption option = CPDF_DefaultConvertOption();
CPDF_StartPDFToHtml(CPDF_TEXT("html.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.html"), option, NULL);

option.page_layout_mode = e_PageLayoutModeBox;
option.html_option = e_CMulTiPagewithBookmark;
CPDF_StartPDFToHtml(CPDF_TEXT("html.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output_multi.html"), option, NULL);
```

4.17 Convert PDF to CSV

Overview

ComPDF Conversion SDK supports converting PDF documents to CSV (Comma-Separated Values). This is commonly used to extract tabular or structured data from PDF documents.

CSV conversion uses the Excel conversion API with `excel_csv_format = true`.

Automatically Create Folders

When multiple CSV files may be output, control whether to automatically create folders through `auto_create_folder`. When this option is enabled, a folder with the same name as the output file will be created in the output path to store the CSV files.

Sample

```
CConvertOption option = CPDF_DefaultConvertOption();
option.excel_csv_format = true;

CPDF_StartPDFToExcel(CPDF_TEXT("csv.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.csv"), option, NULL);

option.excel_worksheet_option = e_CForDocument;
CPDF_StartPDFToExcel(CPDF_TEXT("csv.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output_merged.csv"), option, NULL);
```

4.18 Convert PDF to Image

Overview

ComPDF Conversion SDK provides an API for converting PDF to images.

Setting Image Formats

Supported image formats include:

- JPG
- JPEG
- JPEG2000
- PNG
- BMP
- TIFF
- TGA
- GIF
- WEBP

Setting Image Color Modes

Supported image color modes include:

- `e_cColor`: Color mode, where the image effect is consistent with the original PDF page.
- `e_CGray`: Grayscale mode.
- `e_CBinary`: Black and white mode.

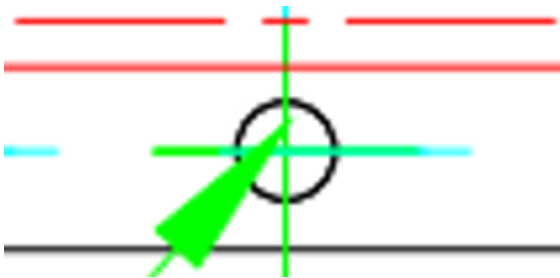
Setting Image Scaling

The SDK supports setting image scaling. If you want to double the image size, set `image_scaling` to `2.0f`; to reduce the image size by half, set `image_scaling` to `0.5f`.

Enhancing Image Path Display

The SDK supports `image_path_enhance` for enhancing the display of image paths. This option can be enabled when you want to improve the display effect of paths within the PDF page.

- Disable `image_path_enhance` option:



- Enable `image_path_enhance` option:



Notice

- A higher `image_scaling` value results in images with higher resolution, but it also increases memory usage and slows down conversion.
- A higher `image_scaling` value does not necessarily mean higher clarity. The clarity also depends on the original image resolution in the document.

Sample

```
CConvertOption option = CPDF_DefaultConvertOption();

option.image_type = e_CJPEG;
CPDF_StartPDFToImage(CPDF_TEXT("jpeg.pdf"), CPDF_TEXT("password"), CPDF_TEXT("path/output"),
option, NULL);

option.image_type = e_CPNG;
option.image_scaling = 2.0f;
CPDF_StartPDFToImage(CPDF_TEXT("png.pdf"), CPDF_TEXT("password"), CPDF_TEXT("path/output"),
option, NULL);
```

4.19 Convert PDF to RTF

Overview

RTF is a popular text format that can retain text format and style data and is convenient for most text readers to read and write.

Sample

```
CConvertOption option = CPDF_DefaultConvertOption();
CPDF_StartPDFToRtf(CPDF_TEXT("rtf.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.rtf"), option, NULL);
```

4.20 Convert PDF to TXT

Overview

When you need to extract text content from a PDF file for data analysis, text mining, or information retrieval, ComPDF Conversion SDK can extract the text into a TXT file.

Preserving Table Format

The SDK supports `txt_table_format` to preserve the table format when writing the TXT file. It is generally recommended to enable this option, especially for data extraction scenarios.

Sample

```
CConvertOption option = CPDF_DefaultConvertOption();
option.txt_table_format = true;

CPDF_StartPDFToTxt(CPDF_TEXT("txt.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.txt"), option, NULL);
```

4.21 Convert PDF to Searchable PDF

Overview

Searchable PDF conversion adds an invisible or visible text layer to an image-based PDF, such as a scanned document, using OCR.

Set Transparent Text Layer

When outputting a searchable PDF, use `transparent_text` to control whether the text layer is transparent.

Sample

```
CPDF_SetDocumentAIModel(CPDF_TEXT("path/documentai.model"), -1);

COCRLanguage languages[] = {e_CENGLISH};

CConvertOption option = CPDF_DefaultConvertOption();
option.enable_ocr = true;
option.languages = languages;
option.language_count = 1;
option.transparent_text = true;

CPDF_StartPDFToSearchablePDF(CPDF_TEXT("scan.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.pdf"), option, NULL);
```

4.22 Convert PDF to OFD

Overview

ComPDF Conversion SDK supports converting PDF documents to OFD documents. Similar to Searchable PDF, OFD conversion also supports OCR, page background preservation, and transparent text layers.

Notice

- If you need to generate searchable OFD output, enable `transparent_text`.
- When `enable_ocr` is enabled, specify the OCR language through `languages`.

Sample

```
CPDF_SetDocumentAIModel(CPDF_TEXT("path/documentai.model"), -1);

COCRLanguage languages[] = {e_CENGLISH};

CConvertOption option = CPDF_DefaultConvertOption();
option.enable_ocr = true;
option.languages = languages;
option.language_count = 1;
option.contain_page_background_image = true;
option.transparent_text = true;

CPDF_StartPDFToofd(CPDF_TEXT("scan.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.ofd"), option, NULL);
```

4.23 Releasing Library Resources

Overview

Release the file and memory resources occupied by the ComPDF Conversion SDK.

Notice

- After calling `CPDF_Release`, the ComPDF Conversion SDK will no longer function properly and must be initialized again before reuse.
- If you only want to release resources occupied by the AI model rather than all SDK resources, call `CPDF_ReleaseDocumentAIModel`.

Sample

```
CPDF_ReleaseDocumentAIModel();  
CPDF_Release();
```

5. Data Extraction Guide

Unleash the power of data with ComPDF Conversion SDK data extraction to detect, recognize, analyze, and extract PDF text, images, tables, and other content.

5.1 Extract PDF to JSON

Overview

Extract text, tables, and images from PDF documents to a JSON file.

Standard Table and Non-standard Table

Tables can commonly be divided into two categories:

- Standard table: The table border and inner lines are complete and clear. There is no need to manually

add table lines to divide table content.

EXTERIOR - RAMPS, DOORS, and LIFTS			
Spec #	Spec	Initials	Notes
101	REMOVAL OF ITEMS FROM WORK AREA Homeowner is responsible for removing objects from the work area. However, contractor is to double check that the items have been removed and that the area is prepared for conversion.		
102	INITIAL INSPECTION AND REPAIR Inspect the general area (concrete, doorway, threshold, etc.) for damage. Immediately report any "unforeseen items" to the Care Manager and Koremen.		
103	PREPARE YARD Prepare the existing yard for the ramp modification and cement. This includes the addition of dirt as needed, the excavating of the land, leveling, and grading of soil away from the home.		
104	DEMOLITION Demolish and remove existing areas <i>only as necessary</i> to construct the modification outlined in the specifications. Dispose of tear out in a code legal dump. Follow all environmental protection measures (lead safe practices) as required by the EPA and local ordinances.		Existing landing and ramps
105A	DOOR THRESHOLD LANDING FOR VPL Construct a preservative treated wood landing at the door threshold height, approximately 42" above ground level or the concrete. 5'x6'. Add new steps. The landing should attach to existing porch and connect to new VPL. (Refer to diagram)		McKinley Collins approved aluminum modular platform and stairs with hand rails.
106	VERTICAL PLATFORM LIFT Install a Bruno (or equivalent) vertical platform lift system per manufacturer's specifications. Ensure the lift has access to appropriate electrical outlets. Lift to have battery backup and call capability from bottom or top.		Pour a concrete base for the VPL. Ensure level. Must have manual override for power outages. Ensure smooth operation.
106a	ELECTRIC FOR INSTALLING VPL Check with local zoning, fire, electric, and other code enforcement agencies regarding contractor licensing required for installation. Install based upon these requirements.		Add outlet to side of house for VPL.
115	CONCRETE LANDING Excavate area to create new concrete landing and loading area. Form and pour 4" thick, 25SF, 2200 PSI concrete slab, to create an area to allow VPL stationary installation and client access point. Use straight, solid forms between temps of 40-100F. All concrete shall be: - o free of voids and cavities. - o treated with liquid curing compound. - o protected from the weather while curing. - o broom finished across direction of traffic. Create a smooth transition onto the driveway/sidewalk from landing. Remove all forms, re-grade and spot seed.		Extend the concrete a minimum 5' wide to the edge of the home.

- Non-standard table: The table lacks borders or clear inner lines, requiring manual additions of table lines to separate content.

features	precision	recall	F1-score	MCC	AUROC
B	0.823 ± 0.311	0.024 ± 0.012	0.047 ± 0.022	0.130 ± 0.055	0.799 ± 0.013
BC	0.818 ± 0.173	0.057 ± 0.016	0.106 ± 0.028	0.197 ± 0.045	0.842 ± 0.005
BT	0.542 ± 0.177	0.051 ± 0.030	0.092 ± 0.052	0.138 ± 0.060	0.786 ± 0.009
BV	0.745 ± 0.040	0.232 ± 0.047	0.350 ± 0.068	0.372 ± 0.055	0.815 ± 0.006
BY	0.781 ± 0.022	0.153 ± 0.014	0.256 ± 0.020	0.314 ± 0.016	0.820 ± 0.004
BTv	0.709 ± 0.011	0.268 ± 0.013	0.389 ± 0.015	0.393 ± 0.013	0.832 ± 0.003
BCTvY	0.793 ± 0.009	0.424 ± 0.011	0.552 ± 0.010	0.541 ± 0.008	0.890 ± 0.002

Table Extraction Option

ComPDF Conversion SDK supports `json_contain_table`. When enabled, the SDK extracts table content from PDFs and outputs the table structure. Otherwise, table content is treated as regular text.

Notice

- Without enabling AI layout analysis or OCR options, tables in the original PDF may not be extracted. It is recommended to enable AI layout analysis or OCR for high-precision table recognition.

Sample

```
CConvertOption option = CPDF_DefaultConvertOption();
option.json_contain_table = true;

CPDF_StartPDFToJson(CPDF_TEXT("json.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.json"), option, NULL);
```

5.2 Extract PDF to Markdown

Overview

Extract text, tables, and images from PDF documents to a Markdown file.

Sample

```
CConvertOption option = CPDF_DefaultConvertOption();

CPDF_StartPDFToMarkdown(CPDF_TEXT("markdown.pdf"), CPDF_TEXT("password"),
CPDF_TEXT("path/output.md"), option, NULL);
```

6. Support

6.1 FAQ

- Does OCR work on x86 architecture?

Currently, OCR only works on x64 architecture.

- Why does my C sample fail on Windows when I pass `"input.pdf"`?

On Windows, most path parameters use `COMPDFKIT_STRING`, which is `const wchar_t*`. Use wide strings such as `L"input.pdf"`, or use the `CPDF_TEXT` helper macro from this guide for string literals.

- Why are my conversion options different from the C++ sample defaults?

`CConvertOption` is a C structure and has no constructor. Initialize it explicitly or use a helper similar to `CPDF_DefaultConvertOption` in this guide.

6.2 Contact Us

Thanks for your interest in ComPDF Conversion SDK, the easy-to-use and powerful development solution. If you encounter technical questions or bug issues when using ComPDF Conversion SDK, please submit the problem report to the [ComPDF team](#). The following information will help us solve your problem:

- ComPDF Conversion SDK product and version.
- Your operating system and IDE version.
- Detailed descriptions of the problem.
- Any other related information, such as an error screenshot.

Contact Information

- Home link: <https://www.compdf.com>
- Technical Support: <https://www.compdf.com/support>
- Email: support@compdf.com

Thanks,

The ComPDF Team